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IIITH Ucchar e-Sudharak: An automatic English pronunciation corrector for school-going children with a teacher in the loop

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Abstract

The language acquisition skills are predominant during childhood. Due to the lack of resources, a gap often arises in the language communication abilities of students. To bridge this gap, it is necessary to consider automatic methods for improving the children language skills. Our IIITH Ucchar e-Sudharak web based tool, deployed in a real environment, helps school-going children to practice English reading skills with a teacher in the loop. The choice of English as the target language is strategic, given its status as a global lingua franca. Within the tool, children are provided with class and chapter-specific English sentences for practice tailored by their teachers. During practice sessions, the tool provides real-time feedback in the form of sentence and word-level scores, which are computed by comparing against teacher (expert) audio recordings. These scores offer insights into word-level pronunciation correctness and sentence-level quality by measuring fluency. Moreover, the tool provides a key feature where teachers can assign specific sentences to individual students and assess their performance. This functionality empowers teachers to personalize their instruction, catering to the specific needs of each student. As far as we know, this is a one-of-a-kind tool specifically designed to meet the needs of school-going children.

Index Terms: Language learning, human-computer interaction, e-learning tool

1. Introduction

Language learning is a fundamental aspect of human development, enabling individuals to communicate, express ideas, and connect with others. Beginning this journey in early childhood is paramount, as it lays the groundwork for effective communication and cognitive development [1]. English, as a global language for communication and commerce, unlocks opportunities in education, careers, and cultural exchange. Therefore, fostering English language skills from a young age equips children to navigate our increasingly interconnected and multicultural world. And, it has shown that online tools are beneficial for learners who may not have convenient access to traditional educational resources [2]. In this demo, a web-based IIITH Ucchar e-Sudharak[1] (Pronunciation e-corrector) tool is developed to assist school children in practicing their English reading skills including teacher involvement. The tool prioritizes teacher involvement, allowing teachers to assign sentences to students, evaluate their performance, and gain insights into their learning progress. Additionally, the students can use the tool for independent practice.

[1]https://asr.iiit.ac.in/LLT

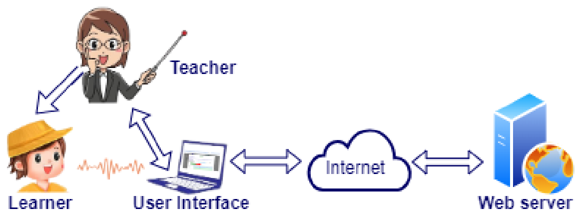

Figure 1: *Proposed architecture of the web-based tool*

2. Proposed architecture

Figure 1 illustrates the architecture of the proposed web-based tool, IIITH Ucchar e-Sudharak. It comprises two main components: a front-end (User Interface) and a back-end (Web server). The front-end is accessible to learners from any location with an internet connection, while the back-end resides on our servers. These components communicate seamlessly over the internet. Learners can access IIITH Ucchar e-Sudharak, that contains practice material tailored by the respective teachers, using a web browser on any electronic device. Further details regarding the front-end and back-end are discussed in the following subsections.

2.1. Front-end functionality

The front-end provides an interactive interface for both learners and teachers. It supports English reading practice for learners and enables teachers to assess learners' reading skills. The primary functionalities include:

2.1.1. Recording Submission

On this view (see Figure 2), all the sentences the learner has chosen for practice will be displayed. Alongside each sentence, there will be two buttons: 'EXPERT' - This button allows the learner to listen to a reference recording of the sentence by a teacher, providing a pronunciation guide. 'RECORD' - Clicking this button will initiate the recording process for the learner. On click of 'RECORD', a pop-up window will appear displaying the sentence to be recorded along with a 'START RECORDING' button. On click of this, the 'START RECORDING' button will automatically be replaced with a 'STOP RECORDING' button. The learner can read the sentence at their own pace and click 'STOP RECORDING' when finished.

2.1.2. Assessment Viewing

Upon submitting their recording (by clicking "STOP RECORDING" button), learners will be presented with an

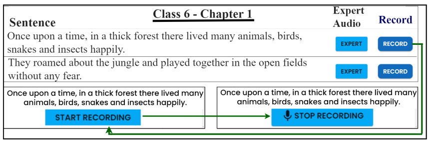

Figure 2: *Recording Submission*

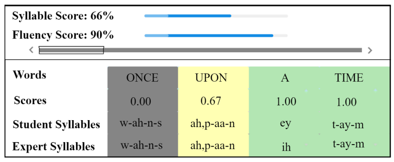

Figure 3: *Assessment Viewing*

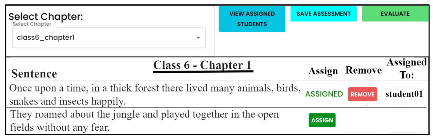

Figure 4: *Assigning sentences to students by teacher*

assessment view. This view provides detailed feedback on their pronunciation (see Figure 4) as follows: **Sentence-level scores -** At the top, the system displays two scores: "Syllable Score" and "Fluency Score". These scores are represented by horizontal bars and indicate the learner's performance as a percentage. **Word-Level pronunciation scores -** Below the sentence-level scores, a grid displays individual words from the recorded sentence. Each word has three associated values: Pronunciation score - reflects the learner's pronunciation score for that particular word. Depending on the score the color indicator will change. Student syllables - This indicates syllables the learner pronounced in the word. Expert syllables - This shows the syllables the teacher pronounced in the word. **Navigation for Long Sentences -** If the sentence is too long to be displayed entirely at once, a horizontal bar with chevrons (greater than/less than symbols) appears at both ends. This allows learners to navigate and review the pronunciation scores for all words in the sentence.

2.1.3. Evaluation of students by teachers

Our tool's key feature allows teachers to add sentences and assign them to students for evaluation as shown in Figure 4. After the teacher clicks on the "EVALUATE" button it brings up a screen similar to Figure 2, but without teacher audio access. The recording and submission process remains the same. Finally, the assessment view shown in Figure 3 appears for teachers to evaluate student's performance.

2.2. Back-end functionality

We performed force alignment using an ASR model trained with the Kaldi toolkit [3]. This process aligns both the learner's recording and the teacher's audio with their corresponding transcripts, estimating the sequence and boundaries of phonemes as well as word boundaries. Using this information and a tool called P2TK [4] the syllables are estimated for both audios. Then the word and sentence level scores are computed as follows: **Word-level pronunciation scores -**The pronunciation score for each phoneme considering goodness-of-pronunciation (GoP) measure described in [5] is utilized. Subsequently, the average score for each word is calculated by averaging the phoneme scores. This score is then referred to as "word-level average score". Similarly, word-level average score is computed for teacher audio and it is used to map the learner word-level score between 0 to 1 by computing the ratio of learner's and teacher's word-level average scores. **Sentence-level scores -** The "Syllable Score" is computed using the formula ((N edit distance)/N) 100, where N represents the count of syllables in the learner's recording. To determine this score, we begin by calculating the edit distance between the learner's and teacher's syllable sequences. Edit distance quantifies the number of edits (insertions, deletions, substitutions) needed to transform one sequence into the other. The sentence-level score is computed similar to the word-level score by averaging the word-level average scores within the entire sentence. Following that, the sentence-level score is mapped between 0 to 1 considering sentence-level score of teacher's audio. This score is referred to as "Fluency Score".

3. Implementation details

To demonstrate the IIITH Ucchar e-Sudharak tool, we selected Class 6 content and divided it into six chapters, each containing around 20 sentences. Initially, all expert audios, which we are referring to as teachers', were synthetically generated using Google's Text-to-Speech (TTS) system. However, the tool eventually leverages recordings from real teachers. We utilized the popular MERN stack (MongoDB, Express.js, React.js, and Node.js) to build both the front-end and back-end of the system, a JavaScript-based framework for developing full-stack web applications. The web server runs on Ubuntu 16.04.7 LTS, a widely used and well-supported Linux operating system. Our tool has been tested successfully in schools around Hyderabad and planning to deploy it soon.

4. Conclusion

We present IIITH Ucchar e-Sudharak, a web-based tool specifically designed to improve children's English reading skills through a teacher-inclusive learning approach. To develop this we have used MERN stack framework. The tool offers feedback at the word and sentence level, comparing learners' pronunciation with that of teachers. Especially for teachers, it allows them to add sentences and evaluate their students. The tool has been successfully tested with students in schools around Hyderabad. We believe that IIITH Ucchar e-Sudharak is a valuable resource that can be easily implemented in classrooms to benefit a wide range of school-going children as well as teachers.

5. References

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